

checkpointer - InMemorySaver

`InMemorySaver` = short-term memory for your agent during a session. It's the quickest way to add multi-turn memory to a LangGraph agent, but it resets every time your program restarts.

Checkpointer

When your LangGraph agent runs, it processes a sequence of steps. A checkpointer saves a snapshot (checkpoint) of the graph state after each step, so you can:

- Resume a conversation from where it left off
- Persist memory across multiple `.invoke()` calls
- Support multi-turn conversations using `thread_id`

Basic Usage

```
from langgraph.checkpoint.memory import InMemorySaver
from langgraph.prebuilt import create_react_agent

checkpointer = InMemorySaver()

agent = create_react_agent(
    model=model,
    tools=tools,
    checkpointer=checkpointer # 📁 attach it here
)

# thread_id groups messages into the same "conversation"
config = {"configurable": {"thread_id": "user_123"}}

# Turn 1
agent.invoke({"messages": [{"role": "user", "content": "My name is Alice"}]}, config)

# Turn 2 – agent still remembers "Alice"
agent.invoke({"messages": [{"role": "user", "content": "What's my name?"}]}, config)
```

thread_id

```
thread_id: "user_123" → [turn 1] → [turn 2] → [turn 3] ✅ remembers context

thread_id: "user_456" → [turn 1] ✅ separate session
```

Meaning that, different `thread_id` = completely separate memory/conversation

Note: the `InMemorySaver` is not good for production, so we can swap it out for the persistent checkpointer like:

```
# PostgreSQL
from langgraph.checkpoint.postgres import PostgresSaver

# SQLite
from langgraph.checkpoint.sqlite import SqliteSaver
```